4.

Between the Galaxies

In the grand tally of cosmic constituents, galaxies are what typically get counted. Latest estimates show that the observable universe may contain a hundred billion of them. Bright and beautiful and packed with stars, galaxies decorate the dark voids of space like cities across a country at night. But just how voidy is the void of space? (How empty is the countryside between cities?) Just because galaxies are in your face, and just because they would have us believe that nothing else matters, the universe may nonetheless contain hard-to-detect things between the galaxies. Maybe those things are more interesting, or more important to the evolution of the universe, than the galaxies themselves.

Our own spiral-shaped galaxy, the Milky Way, is named for its spilled-milk appearance to the unaided eye across Earth's nighttime sky. Indeed, the very word "galaxy" derives from the Greek *galaxias*, "milky." Our pair of nearest-neighbor galaxies, 600,000 light-years distant, are both small and irregularly shaped. Ferdinand Magellan's ship's log identified these cosmic objects during his famous round-the-world voyage of 1519. In his honor, we call them the Large and Small Magellanic Clouds, and they are visible primarily from the Southern Hemisphere as a pair of cloudlike splotches on the sky, parked beyond the stars. The nearest galaxy larger than our own is two million light-years away, beyond the stars that trace the constellation Andromeda. This spiral galaxy, historically dubbed the Great Nebula in Andromeda, is a somewhat more massive and luminous twin of the Milky Way. Notice that the name for each system lacks reference to the existence of stars: Milky Way, Magellanic Clouds, Andromeda Nebula. All three